

Machine Learning (911.236), Statistical Learning Theory (536.505), Advanced Machine Learning (911.936)

Exercise sheet F

Exercise 1.

10 P.

Consider a binary classification problem with training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ where $y_i \in \{-1, +1\}$. AdaBoost constructs an *additive model* of the form

$$F_M(x) = \sum_{m=1}^M \alpha_m h_m(x),$$

where each weak learner $h_m : \mathcal{X} \rightarrow \{-1, +1\}$ is assigned a coefficient $\alpha_m \in \mathbb{R}$. Upon applying the sign function to the additive model, we obtain the final classifier. A statistical interpretation of AdaBoost is obtained by considering the empirical *exponential risk*

$$R(F) = \sum_{i=1}^n \exp(-y_i F(x_i)).$$

At iteration m , AdaBoost updates the current model according to $F_m(x) = F_{m-1}(x) + \alpha_m h_m(x)$ and uses sample weights

$$w_i^{(m)} = \frac{\exp(-y_i F_{m-1}(x_i))}{\sum_{j=1}^n \exp(-y_j F_{m-1}(x_j))}.$$

Tasks:

- (a) Show that at iteration m , AdaBoost can be interpreted as solving the optimization problem

$$(\alpha_m, h_m) = \arg \min_{\alpha, h} R(F_{m-1} + \alpha h),$$

where h ranges over the set of admissible weak learners. Can you explain why this procedure is referred to as *forward stagewise additive modeling*. (2 points)

- (b) Starting from the objective

$$R(F_{m-1} + \alpha h),$$

show that minimizing the exponential risk is equivalent to minimizing

$$\sum_{i=1}^n w_i^{(m)} \exp(-\alpha y_i h(x_i)).$$

Using the fact that $h(x_i) \in \{-1, +1\}$, prove that for fixed h the objective can be written as

$$(1 - \varepsilon_m) e^{-\alpha} + \varepsilon_m e^{\alpha},$$

where

$$\varepsilon_m = \sum_{i=1}^n w_i^{(m)} \mathbf{1}_{\{y_i \neq h(x_i)\}}$$

is the weighted classification error.

Deduce that the optimal coefficient is

$$\alpha_m = \frac{1}{2} \log \left(\frac{1 - \varepsilon_m}{\varepsilon_m} \right).$$

(3 points)

(c) Consider the exponential loss associated with a single training example,

$$\ell_i(F) = \exp(-y_i F(x_i)).$$

Compute the derivative

$$\frac{\partial \ell_i(F)}{\partial F(x_i)}.$$

Show that the AdaBoost weights satisfy

$$w_i^{(m)} \propto \left| \frac{\partial \ell_i(F)}{\partial F(x_i)} \right|_{F=F_{m-1}}.$$

(2 points)

(d) In another variant of boosting, called *gradient boosting*, we construct additive models by fitting weak learners to the negative gradient of a loss function. Compute the negative gradient of the exponential risk

$$R(F) = \sum_{i=1}^n \exp(-y_i F(x_i)).$$

Show that

$$-\frac{\partial R(F)}{\partial F(x_i)} = y_i \exp(-y_i F(x_i)).$$

and explain how AdaBoost can be interpreted as a special instance of gradient boosting and identify the loss function that is implicitly optimized.

(3 points)

Total #points: 10 P.